

Model Risk Specialist — ML/AI Data & Infrastructure Risks (IC5)

Interview Business Case

This is a case-based conversation, not a test with a single right answer. We'll use it as the foundation for a 1-hour live discussion where we can hear your thinking, ask questions, and explore trade-offs together. There's no internal framework you're expected to already know and no hidden "correct" answer we're comparing you against — we care about how you reason under ambiguity, how you prioritize when things compete, and how clearly you communicate the judgment calls you made.

The system at the center of Question 1 is real, working software you can clone, run, and probe yourself — we think that's a more honest test of how you'd actually validate infrastructure and data risk than a purely hypothetical scenario.

Question 1 — Infrastructure & Data Risk Assessment (hands-on)

Background

Solara Digital Bank is a fictional digital bank offering credit cards and personal loans across three countries. Nine months ago, its Platform team launched **Compass**, an internal feature store and serving pipeline: it computes features from transaction, bureau, and app-usage data, and serves them to credit models both in batch (underwriting decisions) and in real time (in-app credit line adjustments).

Compass has become foundational faster than expected — four credit models across two countries now depend on it.

The system you'll be assessing

The codebase behind Compass is available at:
<https://github.com/lennon-cruz/compass-feature-pipeline>.

You're expected to clone it, read the source, and run it yourself: ingest the sample data, compute features through both the batch and streaming paths, and serve a request end to end. The README will walk you through setup. Forming a view on Compass without actually running it will be a noticeably weaker basis for your assessment than testing it directly.

The situation you're stepping into

You've just joined Solara's Model Risk function (second line of defense) as the specialist covering ML/AI data and infrastructure risk. Compass has never had a formal independent risk assessment — no validation, no assigned risk rating, no defined monitoring standard.

At the same time, the Platform and Product teams want to move forward on two changes in the next six weeks:

1. Expand Compass to power a new **instant credit-line top-up** feature that adjusts limits in real time, with no human review of the decision.
2. **Lighten the current change-review process** for feature transformation code, so new features can ship faster.

What you have available

Item	Detail
System name	Compass
System type	Feature store + batch/streaming feature pipeline serving multiple downstream credit models
Primary use	Feature computation and serving for underwriting (batch) and credit-line adjustment (real-time)
Proposed changes	Real-time credit-line top-ups with no human review; lighter review process for feature code changes
Time in production	9 months
Prior formal validation	Informal review at launch only; none since
Monitoring in place	Uptime and latency only — no feature drift, no training/serving parity checks
Human oversight	Engineers periodically eyeball dashboards; no automated alerting
Known incidents	<i>During a routine QA check, someone noticed that a feature value for the same customer</i>

Item	Detail
	<i>looked different depending on which store it was read from, with no obvious explanation. Separately, a support ticket flagged that a customer's credit-line adjustment seemed to reflect an outdated transaction pattern, even though no errors or downtime showed up on any dashboard. Neither was traced to a root cause.</i>
Regulatory/internal context	Solara's regulator requires that credit decisions be explainable and substantiated; internal audit has flagged feature governance as a growing concern as Compass usage has expanded

Your deliverable

Prepare a structured presentation covering:

1. **Risk landscape** — What are the most significant infrastructure and data risks in Compass as it operates today? Focus on what's specific to feature stores and pipelines (training/serving skew, silent staleness, deployment blast radius, unclear data lineage and ownership under a Data Mesh model) rather than generic model risk. Base this on what you actually observe running the system, not just the description above.
 2. **Assessment approach** — If you were running a formal independent review of Compass, what would you actually check? Walk through the tests you'd run against the live system (offline/online feature parity, drift detection, deployment safety and rollback behavior, data quality checks) and how you'd decide whether it's fit for purpose.
 3. **Position on the proposed changes** — Given what you found, what's your independent view on the two proposed changes? Proceed, proceed with conditions, or pause? Be specific about what concerns you most, what would need to be true before you're comfortable, and how you'd communicate that to teams under pressure to move fast.
 4. **Governance foundations** — Regardless of whether the changes proceed, what are the two or three most important controls or monitoring mechanisms Solara should put in place for Compass? What is each meant to catch, and how would you know it's working?
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Question 2 — Governance Framework Design

Solara's (and Nubank's) model and data teams operate under a Data Mesh model — feature and data ownership is decentralized across domain teams with varying levels of risk maturity.

Zooming out from Compass specifically: how would you design or evolve a data governance and control framework for feature stores and model pipelines in this kind of environment?

- What would you standardize across all domains vs. leave to each team's discretion?
 - Which KRIs would you propose at the org level, and how would you keep them meaningful without creating noise?
 - How do you keep governance from becoming a bottleneck to delivery, especially for domains with lower risk maturity?
 - How would you drive adoption across teams that don't report to you?
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What we're assessing

- **Hands-on validation instinct** — did you actually run Compass and let observed behavior inform your judgment, rather than reasoning about it only in the abstract?
- **Risk judgment and prioritization** — can you tell the difference between risks that are material here and ones that are generic or lower-stakes, and explain why?
- **Trade-off reasoning** — do you engage seriously with the tension between shipping fast and governing well, rather than resolving it too easily either way?
- **Framework thinking at scale** — in Question 2, can you design something that works across many teams with uneven maturity, not just for one system?
- **Stakeholder awareness** — does your framing reflect who you're talking to and what a credible independent opinion sounds like to a mixed audience?
- **Clarity of communication** — can you make infrastructure and governance concepts accessible to both technical and non-technical stakeholders?

How to prepare

- Clone and run Compass before reasoning about it. Feed it inputs that exercise both the batch and streaming paths and see where they diverge.
- Think about what's specific to feature-store and pipeline risk — what's different here from validating a single predictive model.
- Structure your reasoning before you structure your slides: start with what you think the core problem is, then work outward to recommendations and evidence.
- Prioritize a clear, well-reasoned position over exhaustive coverage.

- You're welcome to use AI tools in preparing your response — if you do, be transparent about how you used them and what role they played. That transparency is itself something we value.
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Practical details

- **Take-home deadline:** 7 calendar days from receiving this case — please submit your deliverable by end of day.
- **Deliverable format:** your choice — Google Doc, slide deck, or a small repo with a clear README. What matters is that both a technical and a non-technical stakeholder could follow your reasoning.
- **Live session:** 1 hour — 5 min introduction, 35–40 min case walkthrough, 15 min Q&A (yours and ours).
- **Language:** written deliverable in English. The live session can run in Portuguese; we'll ask at least two questions in English during Q&A and ask for your closing recommendation (~2 min) in English.